

SECTION 11 INTERFACING METHODS

This section contains information about the interface logic required to connect the MC68881/MC68882 (FPCP) to an MC68020/MC68030 (MPU) as a coprocessor, or to an MC68000, MC68008, or MC68010 as a peripheral processor.

11.1 FPCP AND MPU INTERFACING

The following paragraphs describe the connecting of the FPCP to an MPU for coprocessor operation using an 8-, 16-, or 32-bit data bus.

11.1.1 32-Bit Data Bus Coprocessor Connection

Figure 11-1 illustrates the coprocessor interface connection of an FPCP to an MPU using a 32-bit data bus. The FPCP is configured to operate over a 32-bit data bus when both the A0 and $\overline{\text{SIZE}}$ pins are connected to VCC.

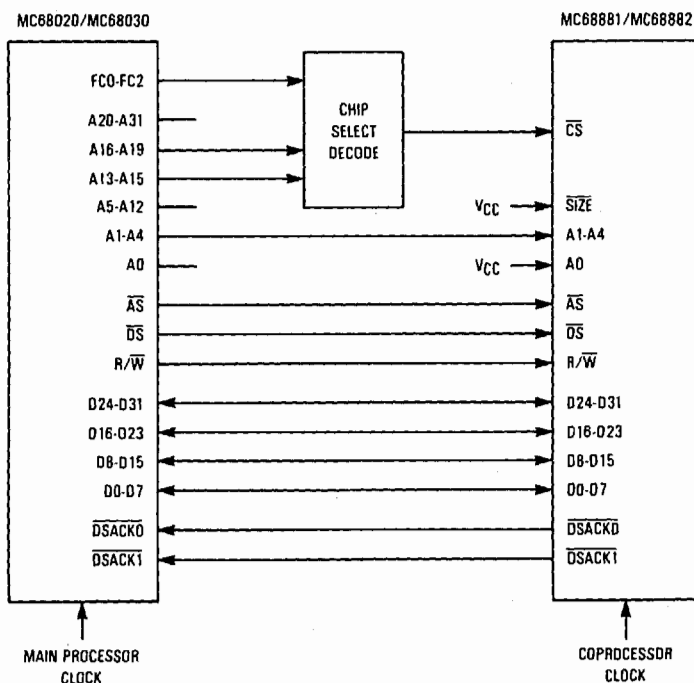


Figure 11-1. 32-Bit Data Bus Coprocessor Connection

Figure 11-2 illustrates the coprocessor interface connection of an FPCP to an MPU using a 16-bit data bus. The FPCP is configured to operate over a 16-bit data bus when the **SIZE** pin is connected to **VCC**, and the **A0** pin is connected to **GND**. The sixteen least significant data pins (**D15–D0**) must be connected to the sixteen most significant data pins (**D31–D16**) when the FPCP is configured to operate over a 16-bit data bus (i.e., connect **D0** to **D16**, **D1** to **D17**, . . . and **D15** to **D31**). The **DSACKx** pins of the two devices are directly connected, although it is not necessary to connect the **DSACK0** pin since the FPCP never asserts it in this configuration.

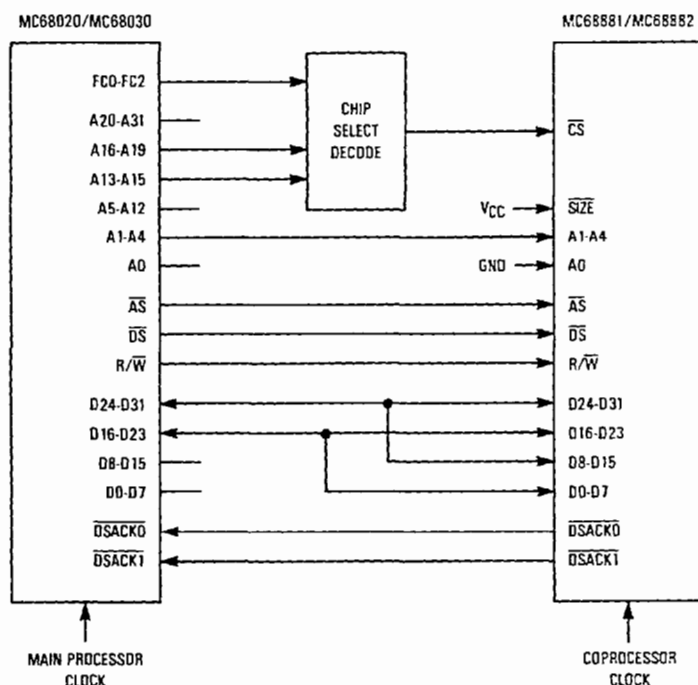


Figure 11-2. 16-Bit Data Bus Coprocessor Connection

11.1.3 8-Bit Data Bus Coprocessor Connection

Figure 11-3 illustrates the connection of an FPCP to an MPU as a coprocessor over an 8-bit data bus. The FPCP is configured to operate over an 8-bit data bus when the **SIZE** pin is connected to **GND**. The 24 least significant data pins (**D23–D0**) must be connected to the eight most significant data pins (**D31–D24**) when the FPCP is configured to operate over an 8-bit data bus (i.e., connect **D0** to **D8**, **D16** and **D24**; **D1** to **D9**, **D17**, and **D25**; . . . and **D7** to **D15**, **D23**, and **D31**). The **DSACKx** pins of the two devices are directly connected, although it is not necessary to connect the **DSACK1** pin since the FPCP never asserts it in this configuration.

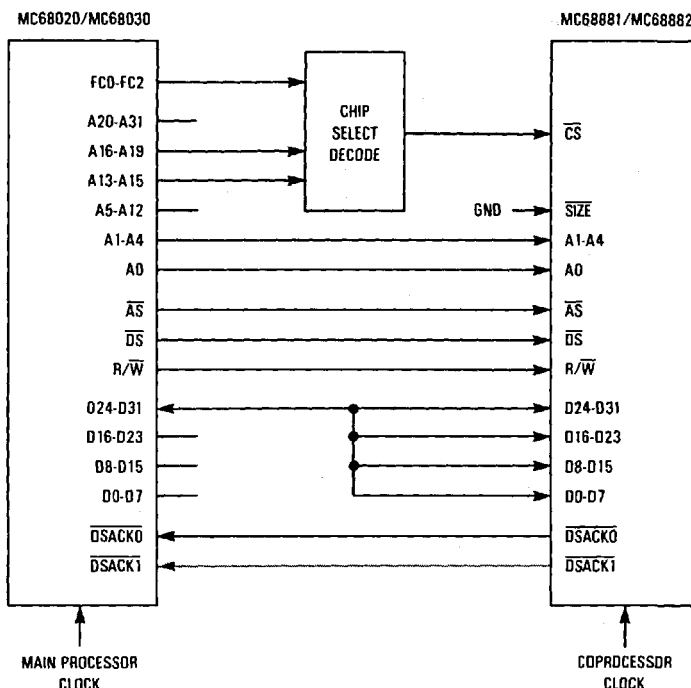


Figure 11-3. 8-Bit Data Bus Coprocessor Connection

11.2 INTERFACING THE FPCP AS A PERIPHERAL

The following paragraphs describe the connecting of the FPCP to an MC68000, MC68008, or MC68010 processor for operation as a peripheral using an 8- or 16-bit data bus.

11.2.1 16-Bit Data Bus Peripheral Processor Connection

Figure 11-4 illustrates the connection of an FPCP to an MC68000 or MC68010 as a peripheral processor over a 16-bit data bus. The FPCP is configured to operate over a 16-bit data bus when the $\overline{SIZE}$ pin is connected to V_{CC} , and the A0 pin is connected to GND. The 16 least significant data pins (D15–D0) must be connected to the 16 most significant data pins (D31–D16) when the FPCP is configured to operate over a 16-bit data bus (i.e., connect D0 to D16, D1 to D17, . . . and D15 to D31). The $\overline{DSACK1}$ pin of the FPCP is connected to the $\overline{DTACK}$ pin of the main processor, and the $\overline{DSACK0}$ pin is not used.

When connected as a peripheral processor, the FPCP chip select ($\overline{CS}$) decode is system dependent. If the MC68000 is used as the main processor, the FPCP $\overline{CS}$ must be decoded in the supervisor or user data spaces. However, if the MC68010 is used for the main processor, the MOVES instruction can be used to emulate any CPU space access that the MPU generates for coprocessor communications. Thus, the $\overline{CS}$ decode logic for such systems may be the same as in an MC68020 or MC68030 system; that is, the FPCP does not use any part of the data address spaces.

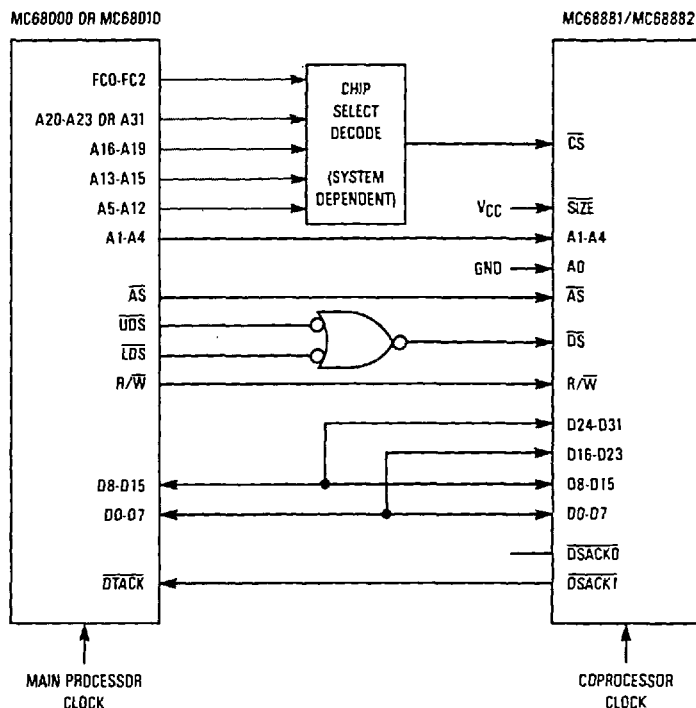


Figure 11-4. 16-Bit Data Bus Peripheral Processor Connection

11.2.2 8-Bit Data Bus Peripheral Processor Connection

Figure 11-5 illustrates the connection of an FPCP to an MC68008 as a peripheral processor over an 8-bit data bus. The FPCP is configured to operate over an 8-bit data bus (i.e., connect D0 to D8, D16, and D24; D1 to D9, D17, and D25; . . . and D7 to D15, D23, and D31). The $\overline{\text{DSACK0}}$ pin of the FPCP is connected to the $\overline{\text{DTACK}}$ pin of the MC68008, and the $\overline{\text{DSACK1}}$ pin is not used.

When connected as a peripheral processor, the FPCP chip-select ($\overline{\text{CS}}$) decode is system dependent, and the $\overline{\text{CS}}$ must be decoded in the supervisor or user data spaces.

11.3 PERIPHERAL PROCESSOR OPERATION

The FPCP can be used as a peripheral processor on systems where the main processor does not have a coprocessor interface by using instruction sequences that emulate the protocol of the coprocessor interface. When an FPCP instruction is encountered by an MC68000, MC68008, or MC68010, the instruction causes an F-line emulator trap to be taken. The trap handler then emulates the coprocessor interface protocol. Refer to **SECTION 7 COPROCESSOR INTERFACE** for details of the communications protocol.

The FPCP requests services from the main processor via coprocessor interface response register primitives. **SECTION 7 COPROCESSOR INTERFACE** describes the main processor service requests required for the execution of each FPCP instruction type. Also included in **SECTION 7 COPROCESSOR INTERFACE** is a summary of all FPCP response primitives.

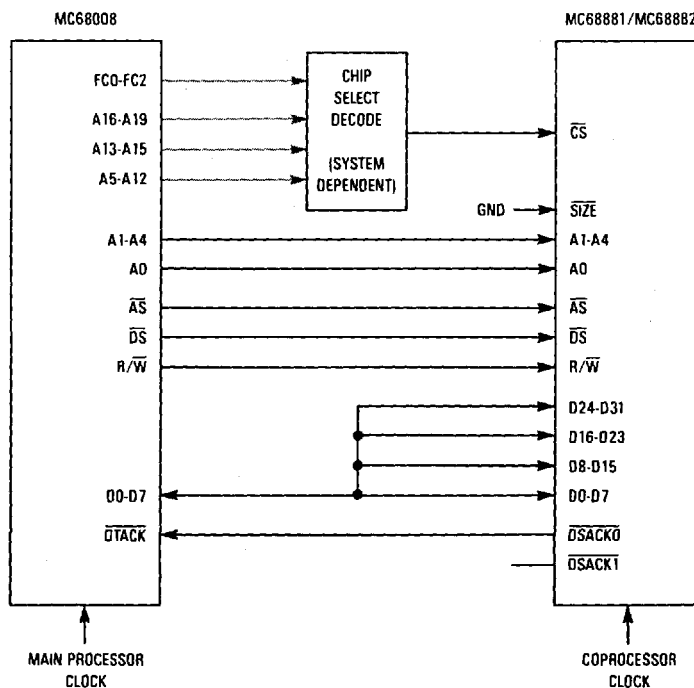


Figure 11-5. 8-Bit Data Bus Peripheral Processor Connection

